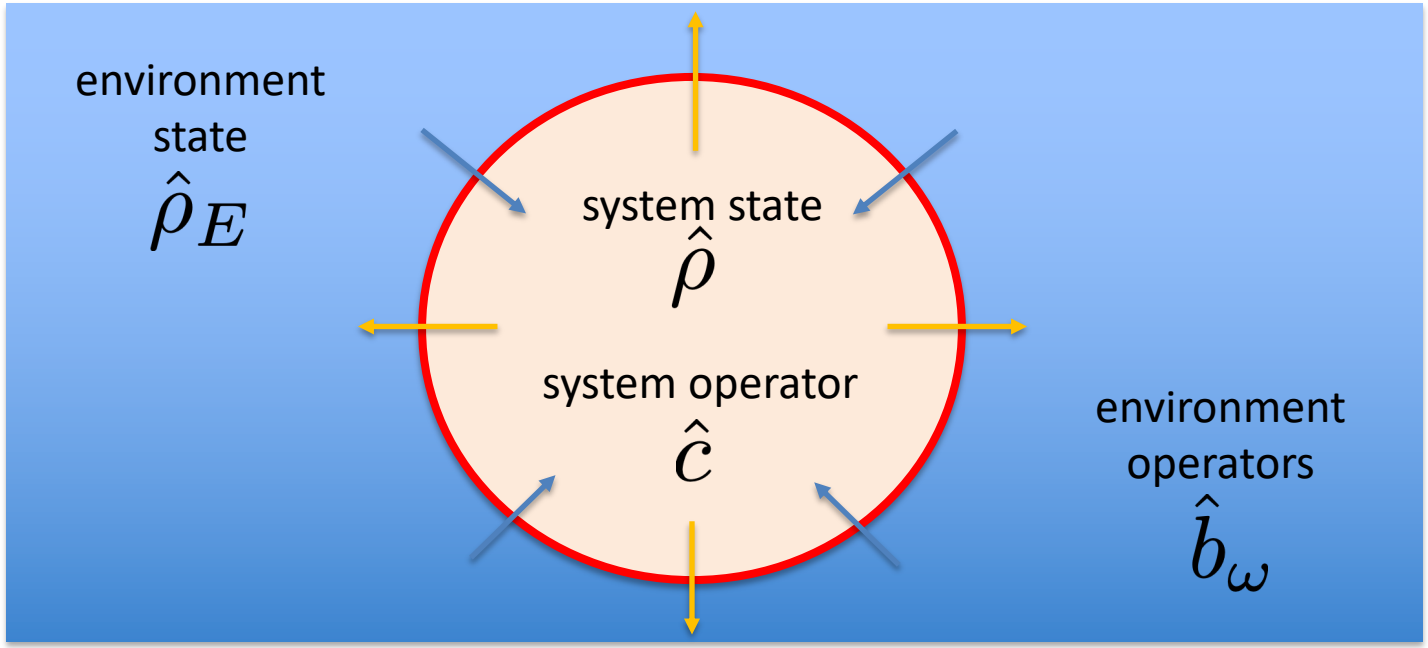
Week 5, Lecture 3
Conditional Quantum Optics

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Office hours: Thursdays 3-4 pm

Bachor & Ralph Ch 6.4, 13.7, 8.3; Nielsen & Chuang, Ch 8.2

Review

What you learned last time…



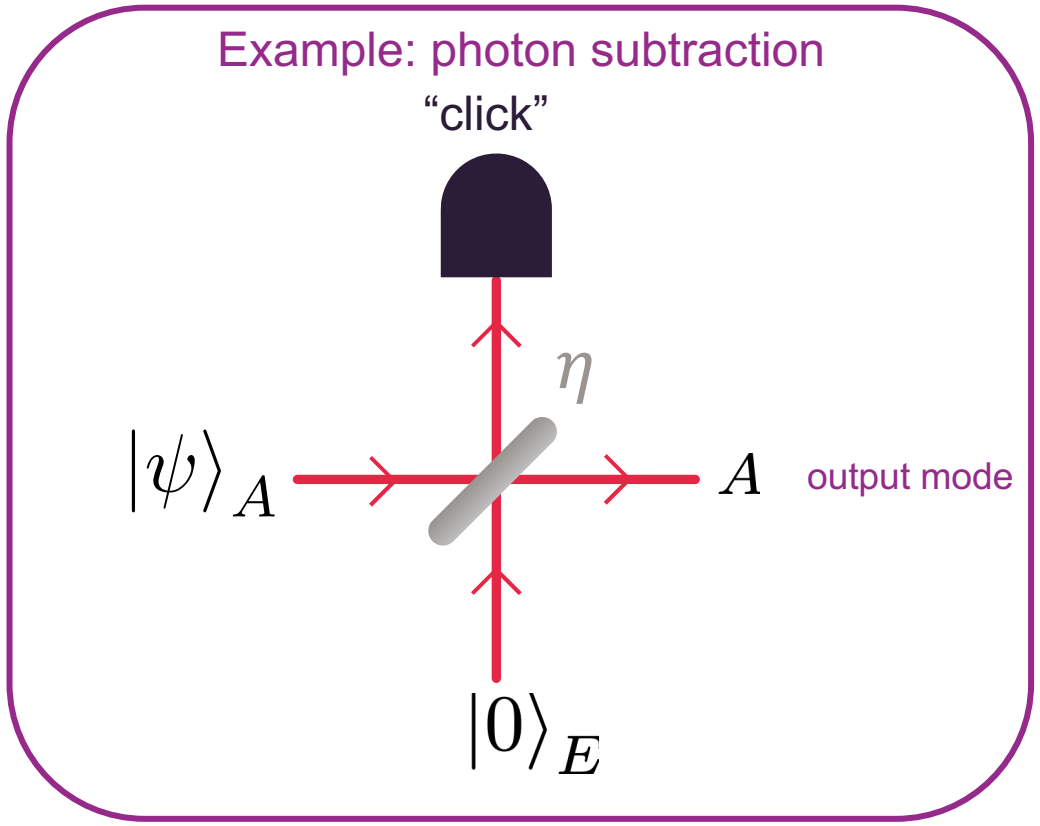
What does the environment do?

How does this evolution develop in time?

Conditional quantum optics

Central question:

What happens when we retain only a particular measurement outcome?



$$\text{input state} \longrightarrow \text{interaction and measurement} \xrightarrow{\text{outcome } k} \text{conditional output state}$$

What you will learn today

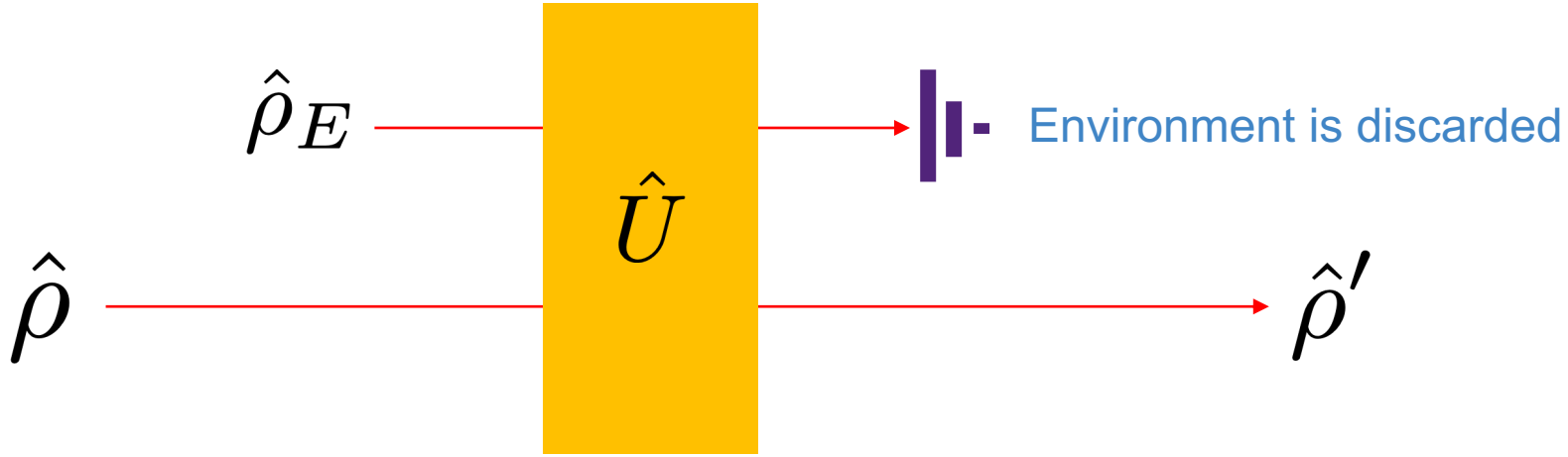
By the end of this lecture, you will be able to:

Explain why conditioning on a measurement outcome produces a **trace-non-increasing quantum operation**.

Calculate the **conditional output state** and its **probability of success**.

Describe how conditional measurements, feedforward and feedback enable **quantum-state engineering and control**.

Review: trace-preserving quantum channels



A system interacts unitarily with an environment:

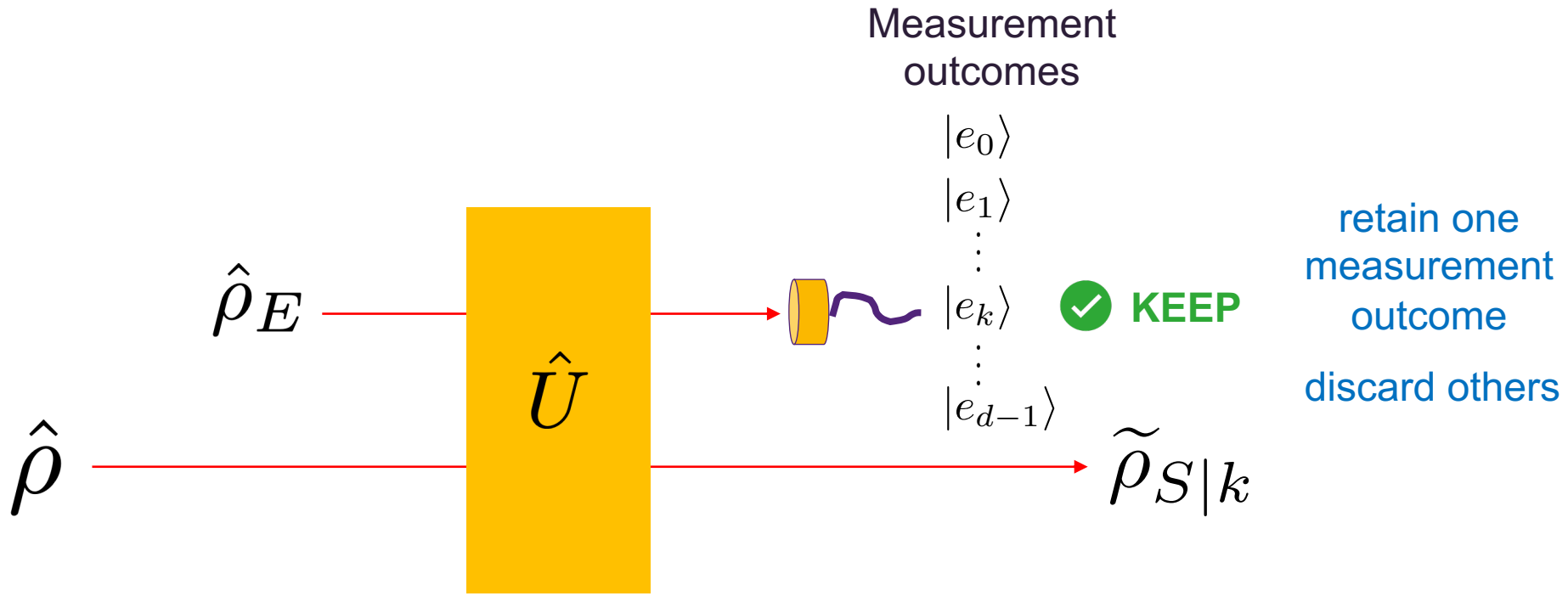
$$\hat{\rho}'_S = \mathcal{E}(\hat{\rho}_S) = \mathrm{Tr}_E \left[\hat{U} \left(\hat{\rho}_S \otimes \hat{\rho}_E \right) \hat{U}^\dagger \right].$$

The output remains normalised:

$$\mathrm{Tr}_S[\hat{\rho}'_S] = 1.$$

All possible environmental outcomes are included.

Conditioning on one measurement outcome



Suppose the environment is measured and we retain only the outcome k, represented by $\hat{\Pi}_k = |e_k\rangle\langle e_k|$.

The corresponding **unnormalised conditional state** of the system is

$$\widetilde{\rho}_{S|k} = \mathrm{Tr}_E \left[\left(\hat{I}_S \otimes \hat{\Pi}_k \right) \hat{U} \left(\hat{\rho}_S \otimes \hat{\rho}_E \right) \hat{U}^\dagger \left(\hat{I}_S \otimes \hat{\Pi}_k \right) \right].$$

Its trace is the probability of obtaining outcome k:

$$p_k = \mathrm{Tr}_S \left[\widetilde{\rho}_{S|k} \right] \leq 1.$$

Normalising the conditional state

The trace of the unnormalised state gives the probability of the selected outcome:

$$p_k = \mathrm{Tr}_S \left[\widetilde{\rho}_{S|k}\right].$$

Provided $p_k > 0$, the normalised conditional state is

$$\rho_{S|k} = \frac{\widetilde{\rho}_{S|k}}{\mathrm{Tr}_S \left[\widetilde{\rho}_{S|k}\right]} = \frac{\widetilde{\rho}_{S|k}}{p_k}.$$

Therefore,

$$\mathrm{Tr}_S \left[\rho_{S|k}\right] = 1.$$

Before normalisation: trace $= p_k$
After normalisation: trace $= 1$

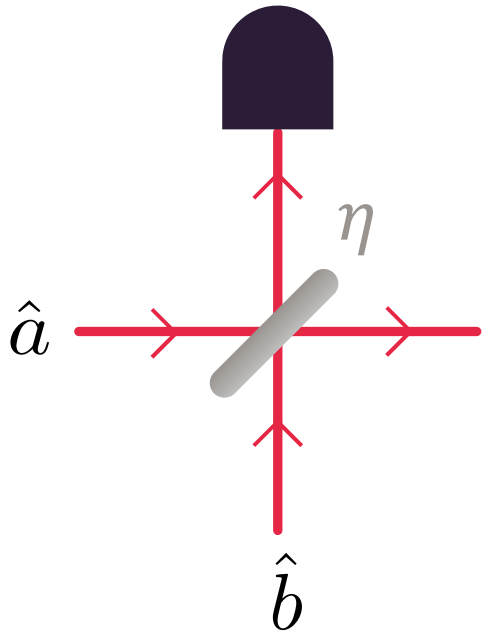
Conditional optical operations

Linear-optical interaction + measurement + post-selection

At the outputs:

- Measure mode $\hat{b}_{out}$ with a photon detector.
- Retain only measurement outcome k.
- The unmeasured mode $\hat{a}_{out}$ is prepared in the conditional state $\rho_{a|k}$.

Outcome k heralds the conditional operation

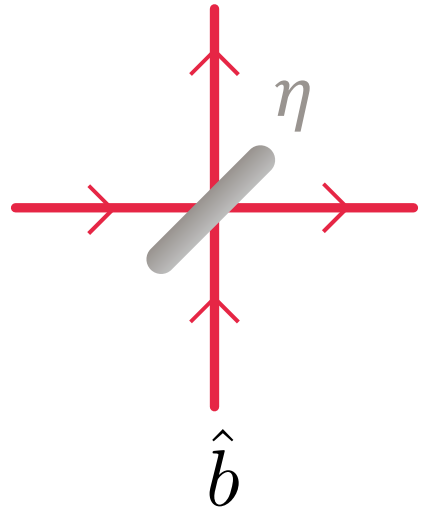


A single photon at a beam splitter

$$|\Psi_{\text{in}}\rangle = (\mu|0\rangle_a + \nu|1\rangle_a) \otimes |0\rangle_b, \qquad |\mu|^2 + |\nu|^2 = 1.$$

For a beam splitter with intensity transmissivity η,

$$|0,0\rangle \longrightarrow |0,0\rangle, \qquad |1,0\rangle \longrightarrow \sqrt{\eta}\,|1,0\rangle + \sqrt{1-\eta}\,|0,1\rangle.$$



Therefore, the joint output state is

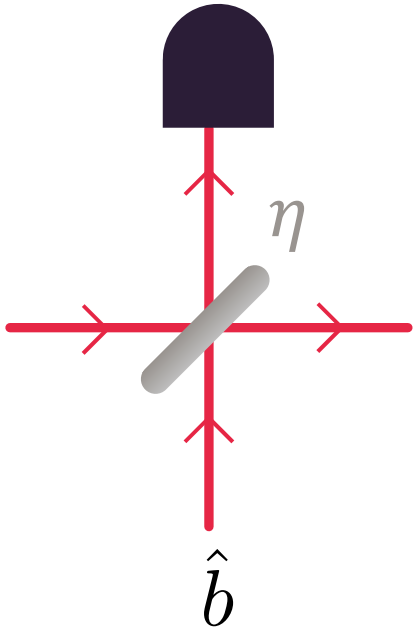
$$|\Psi_{\text{out}}\rangle = \mu|0,0\rangle + \nu\sqrt{\eta}\,|1,0\rangle + \nu\sqrt{1-\eta}\,|0,1\rangle.$$

Same experiment, two descriptions

Write the joint output state in terms of the detector outcomes:

$$|\Psi_{\text{out}}\rangle = \underbrace{(\mu|0\rangle + \nu\sqrt{\eta}|1\rangle)}_{|\phi_0\rangle}|0\rangle_b + \underbrace{\nu\sqrt{1-\eta}|0\rangle}_{|\phi_1\rangle}|1\rangle_b.$$

where $|\phi_0\rangle$ and $|\phi_1\rangle$ are unnormalised branch states.



Ignore the detector outcome

$$\rho_{\text{loss}} = |\phi_0\rangle\langle\phi_0| + |\phi_1\rangle\langle\phi_1|.$$

This is the trace-preserving loss channel studied previously.

Retain the one-photon outcome

$$\widetilde{\rho}_{a|1} = |\phi_1\rangle\langle\phi_1| = |\nu|^2(1-\eta)|0\rangle\langle 0|.$$

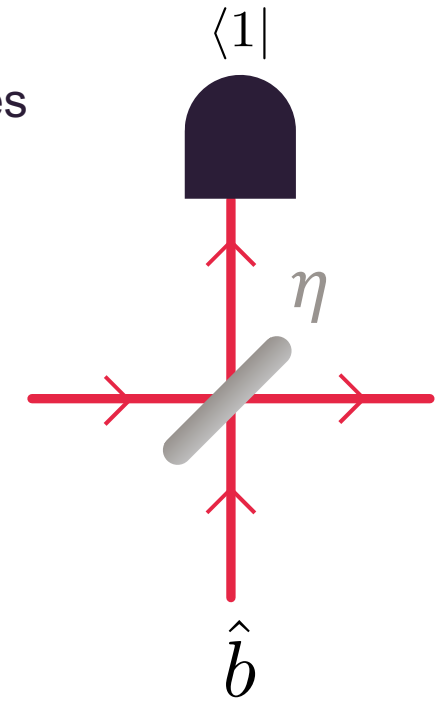
Therefore,

$$p_1 = |\nu|^2(1-\eta), \qquad \rho_{a|1} = |0\rangle\langle 0|.$$

Conditioning on one reflected photon

For an n-photon input and vacuum in the second port, the beam splitter produces

$$|n,0\rangle \longrightarrow \sum_{m=0}^{n} \sqrt{\binom{n}{m}}\, \eta^{(n-m)/2}(1-\eta)^{m/2}|n-m,m\rangle.$$



where m is the number of photons in the detector mode.

Conditioning on exactly one detected photon selects the $m = 1$ term:

$${}_b\langle 1|\hat{U}_{\rm BS}|n,0\rangle = \sqrt{1-\eta}\,\sqrt{n}\,\eta^{(n-1)/2}|n-1\rangle_a.$$

The probability of detecting exactly one photon is

$$p(1|n) = n(1-\eta)\eta^{n-1}.$$

One photon detected $\Longrightarrow$ n photons become n−1 photons.

Photon-subtraction

The result from the previous slide can be written as a single measurement operator:

$$\hat{E}_s = \sqrt{1-\eta}\sum_{n=1}^{\infty}\sqrt{n\,\eta^{n-1}}\,|n-1\rangle\langle n|.$$

The unnormalised photon-subtracted state and success probability are

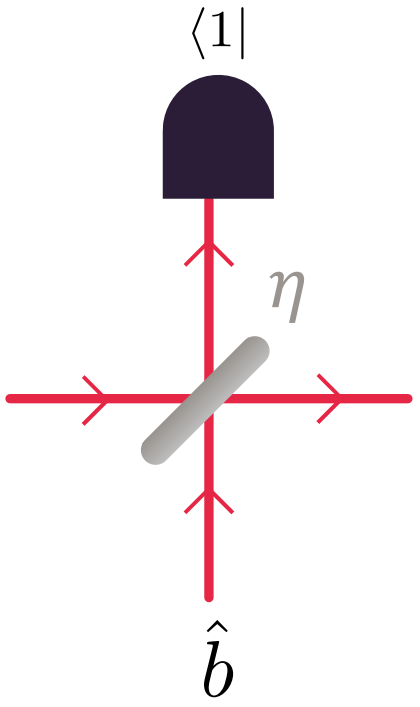
$$\widetilde{\rho}_s = \hat{E}_s\rho\hat{E}_s^\dagger, \qquad p_s = \mathrm{Tr}\left[\hat{E}_s\rho\hat{E}_s^\dagger\right].$$

The normalised conditional state is

$$\rho_s = \frac{\hat{E}_s\rho\hat{E}_s^\dagger}{p_s}.$$

For a highly transmissive beam splitter, $\eta \approx 1$,

$$\hat{E}_s \approx \sqrt{1-\eta}\,\hat{a}.$$



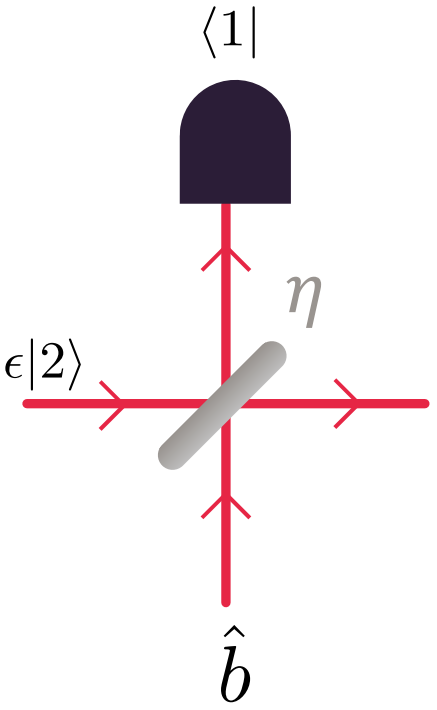
Photon subtraction - example

Consider an input containing up to two photons:

$$|\psi\rangle = \mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle, \qquad |\mu|^2 + |\nu|^2 + |\epsilon|^2 = 1.$$

$$\hat{E}_s|\psi\rangle = \mu\hat{E}_s|0\rangle + \nu\hat{E}_s|1\rangle + \epsilon\hat{E}_s|2\rangle.$$

$|\psi\rangle = \mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle$



Using the photon-subtraction operator from the previous slide,

$$\hat{E}_s|0\rangle = 0,$$

$$\hat{E}_s|1\rangle = \sqrt{1-\eta}\,|0\rangle,$$

$$\hat{E}_s|2\rangle = \sqrt{2\eta(1-\eta)}\,|1\rangle.$$

Substituting these results gives

$$\hat{E}_s|\psi\rangle = \nu\sqrt{1-\eta}\,|0\rangle + \epsilon\sqrt{2\eta(1-\eta)}\,|1\rangle$$
$$= \sqrt{1-\eta}\left(\nu|0\rangle + \sqrt{2\eta}\,\epsilon|1\rangle\right).$$

Photon subtraction - example

$$|\psi\rangle = \mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle, \qquad |\mu|^2 + |\nu|^2 + |\epsilon|^2 = 1.$$

Applying the photon-subtraction operator gives the unnormalized state

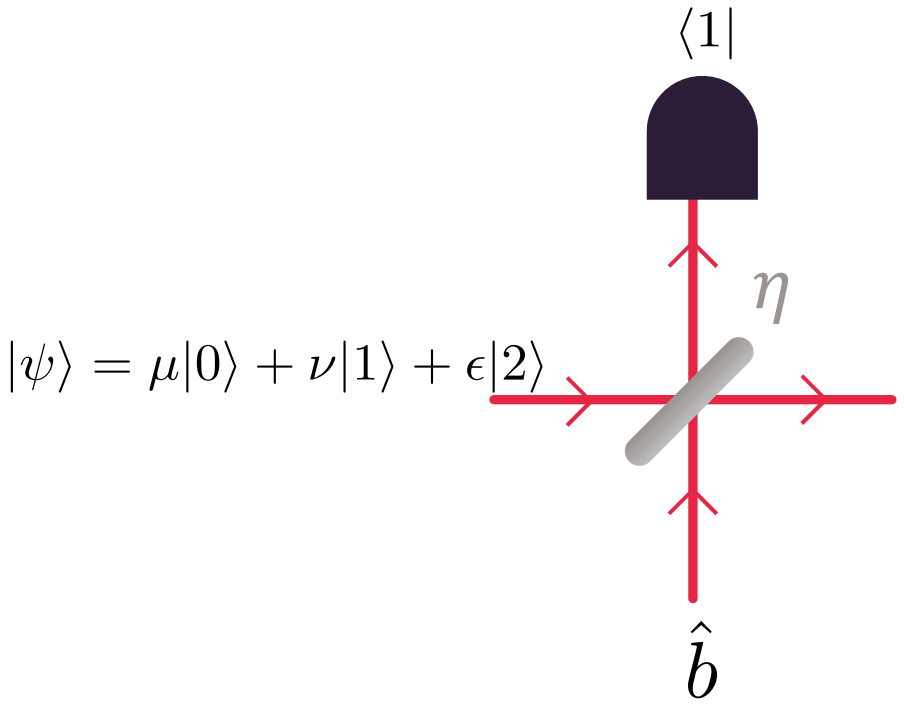
$$\hat{E}_s|\psi\rangle = \sqrt{1-\eta}\left(\nu|0\rangle + \sqrt{2\eta}\,\epsilon|1\rangle\right).$$

The probability of success is

$$p_s = (1-\eta)\left(|\nu|^2 + 2\eta|\epsilon|^2\right).$$

The normalised conditional output is

$$|\psi_s\rangle = \frac{\nu|0\rangle + \sqrt{2\eta}\,\epsilon|1\rangle}{\sqrt{|\nu|^2 + 2\eta|\epsilon|^2}}.$$



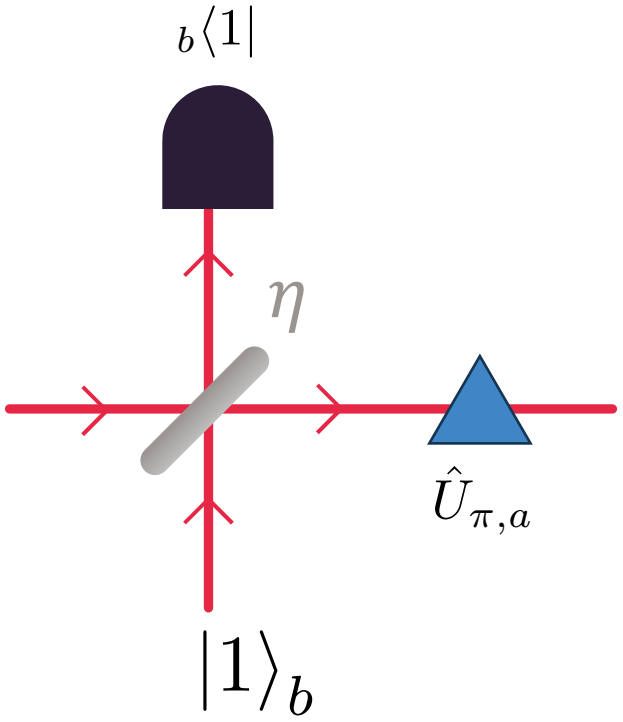
A different conditional operation

Input state

$$|\Psi_{\rm in}\rangle = (\mu|0\rangle_a + \nu|1\rangle_a) \otimes |1\rangle_b, \qquad |\mu|^2 + |\nu|^2 = 1.$$

$$\hat{U}_{\pi,a} = e^{i\pi\hat{n}_a}, \qquad \hat{U}_{\pi,a}|n\rangle_a = (-1)^n|n\rangle_a.$$

$$|\psi\rangle = \mu|0\rangle + \nu|1\rangle$$



Mathematical model of the entire circuit

$$|\widetilde{\phi}_c\rangle_a = \hat{U}_{\pi,a}\,{}_b\langle 1|\,\hat{U}_{\rm BS}(\eta)\,|\Psi_{\rm in}\rangle.$$

Calculate the output selected by the detector

Beam-splitter convention

For intensity transmissivity η, choose

$$\hat{a}_{\rm in}^\dagger \longrightarrow \sqrt{\eta}\,\hat{a}^\dagger + \sqrt{1-\eta}\,\hat{b}^\dagger, \qquad \hat{b}_{\rm in}^\dagger \longrightarrow -\sqrt{1-\eta}\,\hat{a}^\dagger + \sqrt{\eta}\,\hat{b}^\dagger.$$

Vacuum component of the signal

$$|0,1\rangle \xrightarrow{\hat{U}_{\rm BS}} -\sqrt{1-\eta}\,|1,0\rangle + \sqrt{\eta}\,|0,1\rangle \xrightarrow{{}_b\langle 1|} \sqrt{\eta}\,|0\rangle_a.$$

One-photon component of the signal

$$\begin{aligned} |1,1\rangle \xrightarrow{\hat{U}_{\rm BS}} &-\sqrt{2\eta(1-\eta)}\,|2,0\rangle \\ &+ (2\eta-1)|1,1\rangle \\ &+ \sqrt{2\eta(1-\eta)}\,|0,2\rangle. \end{aligned}$$

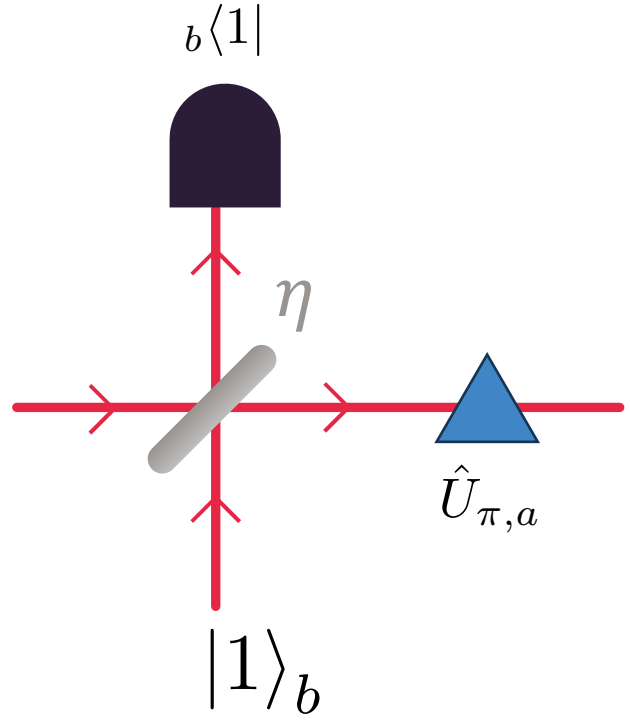
Only the middle term contains one photon in detector mode b:

$$|1,1\rangle \xrightarrow{{}_b\langle 1|} (2\eta-1)|1\rangle_a \xrightarrow{\hat{U}_\pi} (1-2\eta)|1\rangle_a.$$

Combining the two input components gives

$$\boxed{|\widetilde{\phi}_c\rangle_a = \mu\sqrt{\eta}\,|0\rangle_a + \nu(1-2\eta)|1\rangle_a}$$

$|\psi\rangle = \mu|0\rangle + \nu|1\rangle$



The one-photon amplitude is amplified relative to vacuum

From the conditional output derived on the previous slide:

$$|\widetilde{\phi}_c\rangle = \mu\sqrt{\eta}\,|0\rangle + \nu(1-2\eta)|1\rangle.$$

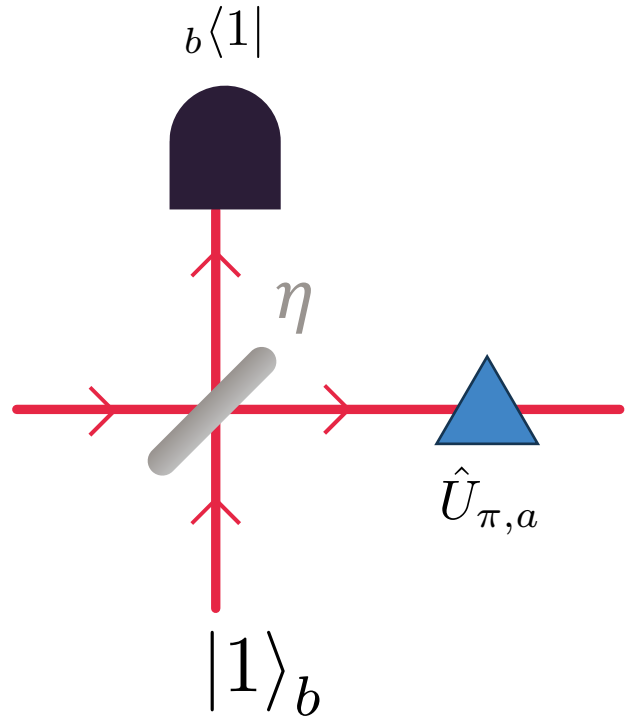
Factor out the common amplitude $\sqrt{\eta}$:

$$|\widetilde{\phi}_c\rangle = \sqrt{\eta}\,(\mu|0\rangle + g\nu|1\rangle)\,, \qquad \boxed{g = \frac{1-2\eta}{\sqrt{\eta}}}.$$

$$\frac{\text{one-photon amplitude}}{\text{vacuum amplitude}}: \qquad \frac{\nu}{\mu} \longrightarrow g\,\frac{\nu}{\mu}.$$

Normalised conditional state $\boxed{|\phi_c\rangle = \dfrac{\mu|0\rangle + g\nu|1\rangle}{\sqrt{|\mu|^2 + g^2|\nu|^2}}}$

The circuit provides amplification when $\quad g > 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{1}{4}.$

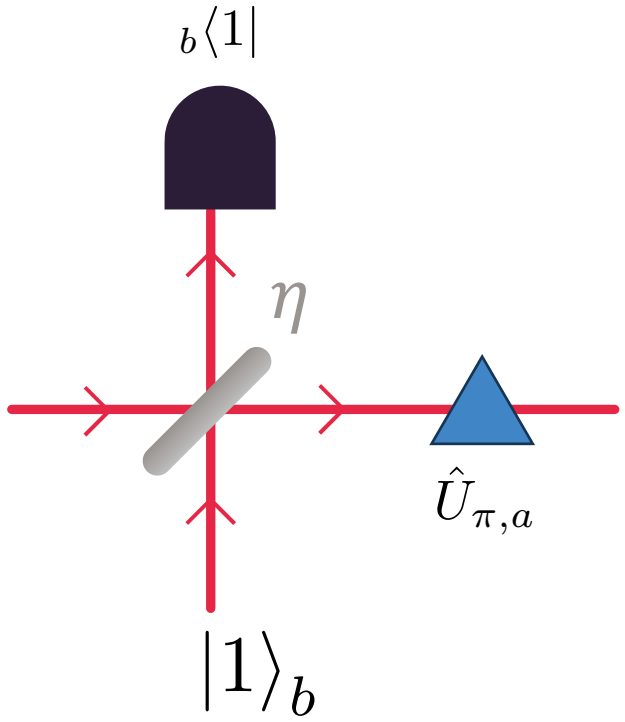


Noiseless amplification

Small weak-coherent-state

For a sufficiently weak coherent state,

$$|\alpha\rangle \simeq |0\rangle + \alpha|1\rangle \longrightarrow |0\rangle + g\alpha|1\rangle \simeq |g\alpha\rangle, \qquad g|\alpha| \ll 1.$$



Noiseless amplification

Deterministic phase-insensitive amplification

A deterministic amplifier with amplitude gain $g > 1$ must introduce an additional mode:

$$\hat{a}_{\text{out}} = g\hat{a}_{\text{in}} + \sqrt{g^2 - 1}\,\hat{v}^\dagger.$$

always succeeds, but must add noise

where $\hat{v}$ is an independent noise mode.

Conditional noiseless amplification

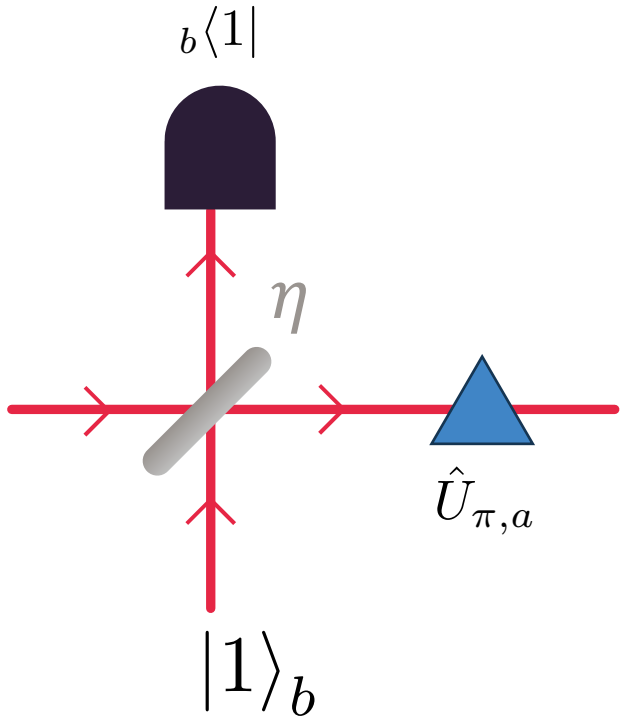
For the selected detector outcome, our circuit produces

$$\mu|0\rangle + \nu|1\rangle \xrightarrow{\text{detect } 1} |\phi_c\rangle \propto \mu|0\rangle + g\nu|1\rangle.$$

no additional noise, but does not always succeed

Probability of success $p_{\text{success}} = \langle\widetilde{\phi}_c|\widetilde{\phi}_c\rangle$

$$= \eta|\mu|^2 + (1-2\eta)^2|\nu|^2$$
$$= \eta\left(|\mu|^2 + g^2|\nu|^2\right).$$



A two-stage conditional optical circuit

Input state

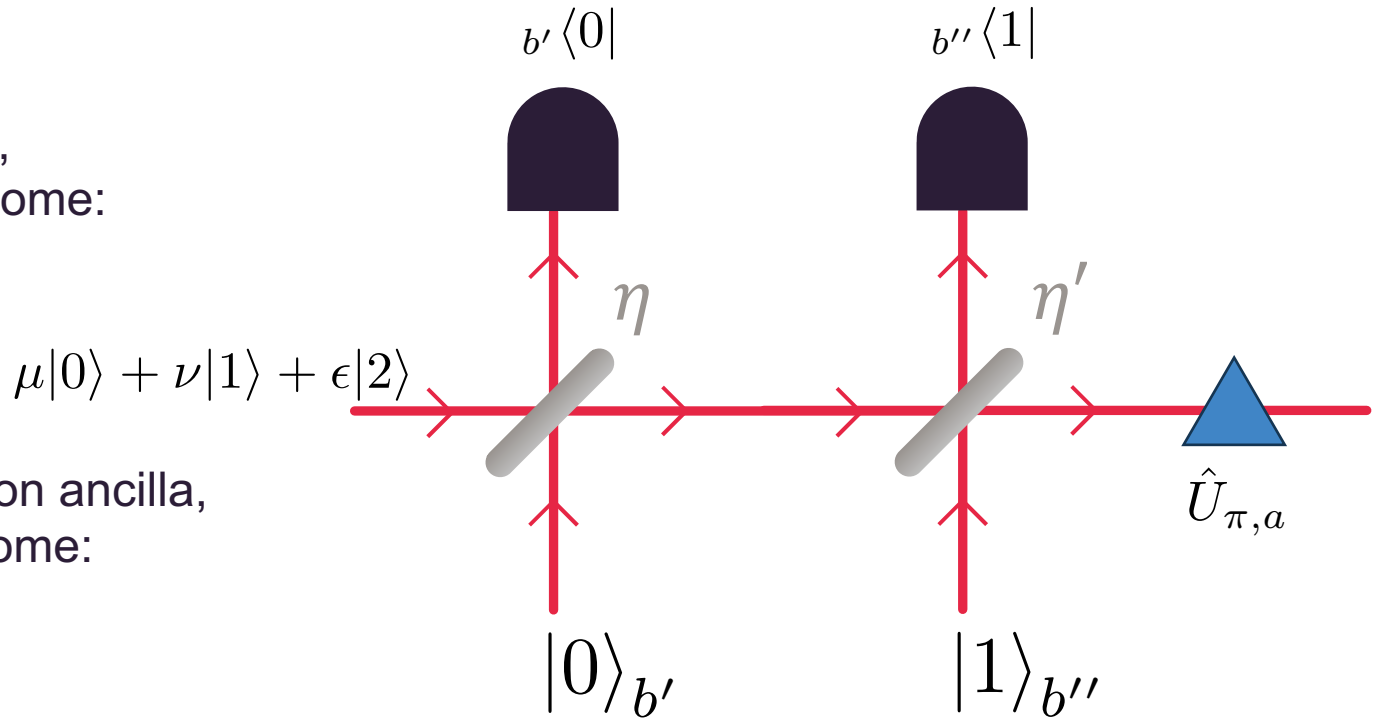
$$|\psi_{\rm in}\rangle_a = \mu|0\rangle_a + \nu|1\rangle_a + \epsilon|2\rangle_a.$$

The first beam splitter uses a vacuum ancilla, followed by selection of the zero-photon outcome:

$$\hat{M}_0(\eta) = {}_{b'}\langle 0|\, \hat{U}_{\rm BS}^{(ab')}(\eta)\, |0\rangle_{b'}.$$

The second beam splitter uses a single-photon ancilla, followed by selection of the one-photon outcome:

$$\hat{M}_1(\eta') = {}_{b''}\langle 1|\, \hat{U}_{\rm BS}^{(ab'')}(\eta')\, |1\rangle_{b''}.$$



The final phase shift is $\hat{U}_{\pi,a} = e^{i\pi\hat{n}_a}$.

$$\boxed{|\widetilde{\phi}_c\rangle_a = \hat{U}_{\pi,a}\, \hat{M}_1(\eta')\, \hat{M}_0(\eta)\, |\psi_{\rm in}\rangle_a}$$

The circuit flips only the two-photon phase

$$|0\rangle \longrightarrow \sqrt{\eta'}\,|0\rangle,$$
$$|1\rangle \longrightarrow \sqrt{\eta}(1-2\eta')|1\rangle,$$
$$|2\rangle \longrightarrow \eta\sqrt{\eta'}(3\eta'-2)|2\rangle.$$

We want the first two factors to be equal and the third to have the opposite sign:

$$\sqrt{\eta}(1-2\eta') = \sqrt{\eta'}, \qquad \eta(3\eta'-2) = -1.$$

The solution is

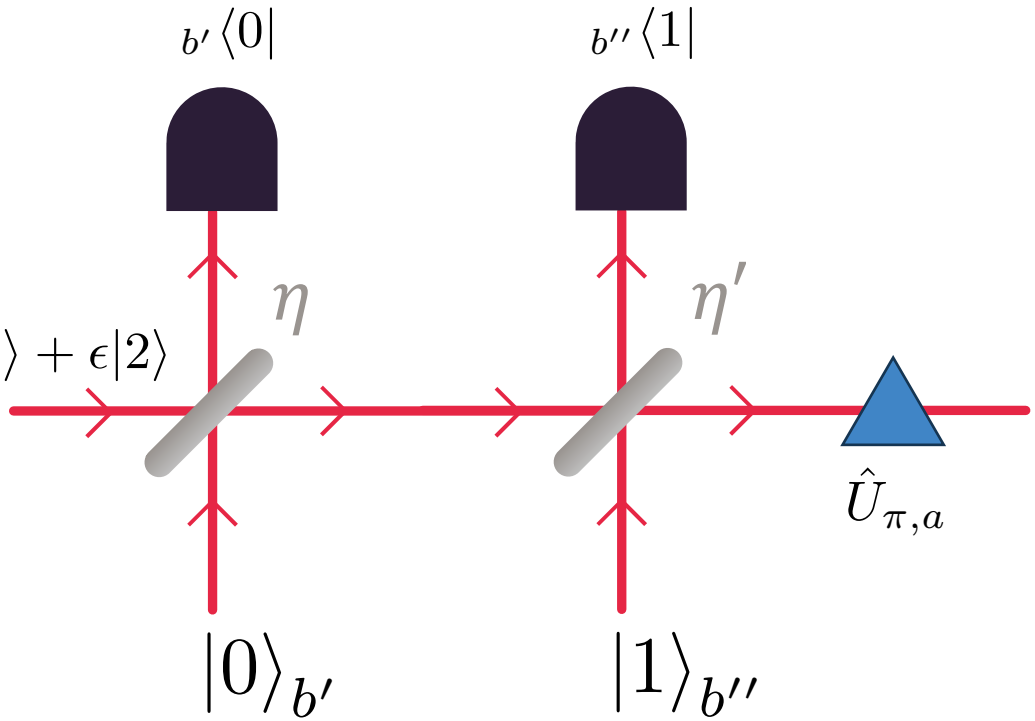
$$\eta = 5-3\sqrt{2}, \qquad \eta' = \frac{3-\sqrt{2}}{7}.$$

$$\boxed{|\widetilde{\phi}_c\rangle = \sqrt{\eta'}\,(\mu|0\rangle + \nu|1\rangle - \epsilon|2\rangle)}$$

After normalisation,

$$\boxed{\mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle \longrightarrow \mu|0\rangle + \nu|1\rangle - \epsilon|2\rangle}$$

$$p_{\text{success}} = \eta' = \frac{3-\sqrt{2}}{7} \approx 0.227.$$

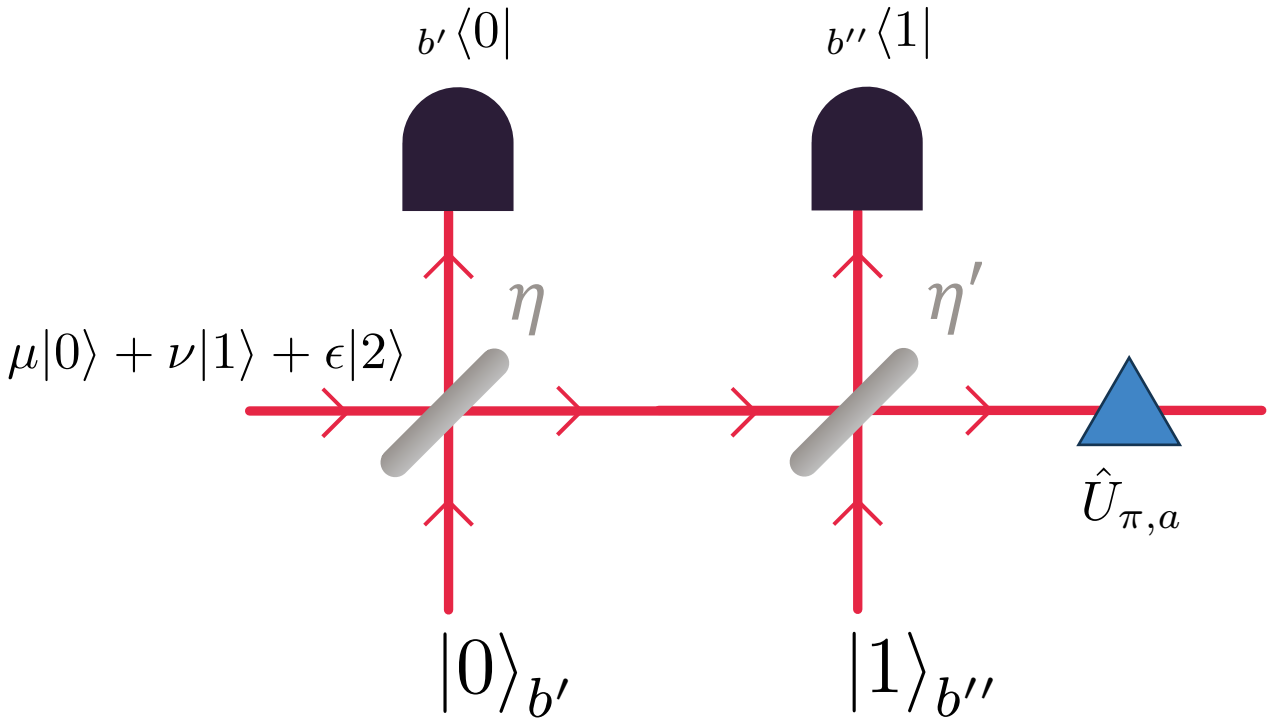


Nonlinear sign-shift gate

$$\mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle \longrightarrow \mu|0\rangle + \nu|1\rangle - \epsilon|2\rangle$$

Why is this useful?

- This is a **nonlinear sign-shift gate**.
- It creates an effective optical nonlinearity using measurement.
- It can be used to construct entangling gates for linear-optical quantum computing.



Noiseless amplification cannot be deterministic

$$|\psi_\pm\rangle = \frac{|0\rangle \pm g^{-1}|1\rangle}{\sqrt{1+g^{-2}}}, \qquad g > 1.$$

Their overlap is nonzero: $\langle\psi_-|\psi_+\rangle = \frac{1-g^{-2}}{1+g^{-2}} > 0.$

Apply ideal noiseless amplification
Since the one-photon amplitude is multiplied by g,

$$|\psi_\pm\rangle \longrightarrow |\psi'_\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}.$$

The output states are orthogonal: $\boxed{\langle\psi'_-|\psi'_+\rangle = 0}$

A deterministic quantum operation cannot transform nonorthogonal states into perfectly distinguishable orthogonal states

What about the nonlinear sign-shift gate?

Consider two arbitrary states in the zero-, one- and two-photon subspace:

$$|\psi\rangle = \mu|0\rangle + \nu|1\rangle + \epsilon|2\rangle,$$
$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle + \delta|2\rangle.$$

Their initial overlap is

$$\langle\psi|\phi\rangle = \mu^*\alpha + \nu^*\beta + \epsilon^*\delta.$$

The nonlinear sign shift is deterministic in principle. Only its measurement-based linear-optical implementation is probabilistic.

Apply the nonlinear sign shift

$$|\psi'\rangle = \mu|0\rangle + \nu|1\rangle - \epsilon|2\rangle,$$
$$|\phi'\rangle = \alpha|0\rangle + \beta|1\rangle - \delta|2\rangle.$$

but producing a strong, low-noise nonlinearity at the single-photon level is experimentally difficult.

The overlap is unchanged:

$$\boxed{\langle\psi'|\phi'\rangle = \mu^*\alpha + \nu^*\beta + \epsilon^*\delta = \langle\psi|\phi\rangle}$$

This transformation is generated by the unitary $\hat{U}_{\rm NS} = \exp\left[i\frac{\pi}{2}\hat{n}(\hat{n}-1)\right]$, $\hat{U}_{\rm NS}^\dagger\hat{U}_{\rm NS} = \hat{I}$.

Summary

Conditional measurements produce probabilistic optical operations: the unnormalised output contains the success probability and is normalised only after the selected detector outcome occurs.

Conditional operations reshape photon-number amplitudes: noiseless amplification increases the one-photon amplitude relative to vacuum.

Conditional measurement can mimic an optical nonlinearity: the sign-shift operation reverses the phase of $|2\rangle$ while leaving $|0\rangle$ and $|1\rangle$ unchanged; it is unitary in principle, but its linear-optical implementation is probabilistic.